CONGRUENTIAL EULER NUMBERS: A FORMAL PROOF OF THE ODD-PRIME FINITE-DIFFERENCE THEOREM

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ABSTRACT. Let p be an odd prime, $m \mid p-1$, $N = mp$, and $h = (p-1)/m$. For all $0 \le j < N$ we prove

$$v_p\Big(\Delta_{uh}^k E_{Nn}^{(N,j)}\Big) \ge \min\{k, mn\} \qquad (u \ge 1).$$

This proves every odd-prime instance of Nishibuchi's Conjecture 3.9, with the explicit sufficient threshold $n_0 = \lceil r/m \rceil$, and also gives a stronger displacement-sensitive modulus. The proof is presented as a complete chain of identities, lemmas, and coefficient extractions. Exact computations are included only as independent checks of each proof interface. The 2-adic branch is not treated.

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Date: June 2026.

2020 *Mathematics Subject Classification.* 11B68, 11B83, 11S80, 33C20.

Key words and phrases. Congruential Euler numbers, generalized Euler numbers, p-adic congruences, finite differences, hypergeometric sections, Mahler expansions.

1. Definitions and principal result

Definition 1.1 (Congruential Euler numbers)**.** For $N \geq 1$ and $0 \leq j < N$, define

$$H_{N,j}(t) = \sum_{n \geq 0} \frac{t^{Nn+j}}{(Nn+j)!}, \tag{1.1}$$

$$F_{N,j}(t) = \sum_{n \geq 0} E_{Nn}^{(N,j)} \frac{t^{Nn}}{(Nn)!}, \tag{1.2}$$

by the identity

$$F_{N,j}(t) H_{N,j}(t) = t^j. \tag{1.3}$$

Equivalently,

$$H_{N,j}(t) = \frac{1}{N} \sum_{\ell=0}^{N-1} \zeta_N^{-j\ell} e^{\zeta_N^{\ell} t}. \tag{1.4}$$

Proposition 1.2 (Triangular recurrence)**.** *For every $n \geq 0$,*

$$\sum_{q=0}^{n} \binom{Nn+j}{Nq} E_{Nq}^{(N,j)} = \begin{cases} j!, & n = 0, \\ 0, & n > 0. \end{cases} \tag{1.5}$$

Proof. Multiply the two divided-power series in (1.3). The coefficient of $t^{Nn+j}/(Nn+j)!$ is the left side of (1.5); the right side t^j contributes $j!$ only when $n = 0$. □

Conjecture 1.3 (Nishibuchi, Conjecture 3.9)**.** *Following* [3, Conjecture 3.9]*, let p be prime, suppose $m \mid p-1$ or $m = 2$, put $q_0 = \operatorname{lcm}(2, p-1)$, and let $0 \leq j < mp$. Then $E_{mpn}^{(mp,j)} \in \mathbb{Z}_p$, and for every $r \geq 1$ there is an n_0 such that*

$$E_{mpn+q_0 p^r}^{(mp,j)} \equiv E_{mpn}^{(mp,j)} \pmod{p^r} \qquad (n \geq n_0).$$

Theorem 1.4 (Finite-difference theorem)**.** *Let p be an odd prime, let $m \mid p-1$, and put*

$$N = mp, \qquad h = \frac{p-1}{m}, \qquad e_n^{(N,j)} = E_{Nn}^{(N,j)}.$$

For $d \geq 1$, write $(\Delta_d a)_n = a_{n+d} - a_n$. Then, for every $0 \leq j < N$ and $k, n \geq 0$,

$$\boxed{v_p\Big(\Delta_h^k e_n^{(N,j)}\Big) \geq \min\{k, mn\}.} \tag{1.6}$$

More generally, for every $u \geq 1$,

$$\boxed{v_p\Big(\Delta_{uh}^k e_n^{(N,j)}\Big) \geq \min\{k, mn\}.} \tag{1.7}$$

In particular, every $e_n^{(N,j)}$ lies in $\mathbb{Z}_p$.

Theorem 1.5 (Displacement-sensitive modulus)**.** *Under the hypotheses of Theorem 1.4, for every $q \geq 1$,*

$$\boxed{v_p\Big(e_{n+hq}^{(N,j)} - e_n^{(N,j)}\Big) \geq \min\{mn, 1 + v_p(q)\}.} \tag{1.8}$$

Consequently, for $r, u \geq 1$,

$$e_{n+uhp^{r-1}}^{(N,j)} \equiv e_n^{(N,j)} \pmod{p^{\min\{mn, r+v_p(u)\}}}. \tag{1.9}$$

In the original Euler-number index,

$$\boxed{E_{mpn+u(p-1)p^r}^{(mp,j)} \equiv E_{mpn}^{(mp,j)} \pmod{p^{\min\{mn, r+v_p(u)\}}}.} \tag{1.10}$$

Corollary 1.6 (Answer to the conjecture for odd primes)**.** *Every odd-prime instance of Conjecture 1.3 is true. A uniform sufficient starting point is*

$$\boxed{n_0 = \left\lceil \frac{r}{m} \right\rceil},$$

and the original-index displacement is $(p-1)p^r$*. The result is strictly stronger than the conjecture because it proves all higher finite differences* (1.6) *and the local modulus* (1.8)*. No assertion is made here for* $p = 2$*.*

2. Hurwitz–Dwork translation

Put $\gamma_d = t^d/d!$. For a characteristic-zero ring R, let

$$\mathcal{H}(R) = \left\{ \sum_{d\geq 0} a_d \gamma_d : a_d \in R \right\}, \qquad \gamma_r \gamma_s = \binom{r+s}{r} \gamma_{r+s}.$$

Let $\mathsf{U} = d/dt$ and $\mathsf{E} = t\mathsf{U}$. Then

$$\mathsf{U}\gamma_d = \gamma_{d-1}, \qquad \mathsf{E}\gamma_d = d\gamma_d.$$

Fix p, m, N, h as in Theorem 1.4, and define

$$D = \mathsf{U}^p, \qquad \mathscr{A} = D^{p-1} - \mathsf{I} = \mathsf{U}^{Nh} - \mathsf{I}, \qquad \delta = D^p - D = D\mathscr{A}. \tag{2.1}$$

Lemma 2.1 (Operator dictionary)**.** *If* $X = \sum_{n\geq 0} x_n \gamma_{Nn}$*, then*

$$\mathscr{A}^k X = \sum_{n\geq 0} \Delta_h^k x_n \gamma_{Nn}, \tag{2.2}$$

$$\delta^s X = \sum_{mn\geq s} \Delta_h^s x_n \gamma_{Nn-ps}. \tag{2.3}$$

Proof. Since $Nh = p(p-1)$,

$$\mathsf{U}^{Nh} X = \sum_{n\geq 0} x_{n+h} \gamma_{Nn}.$$

Thus $\mathscr{A}$ acts as Δ_h on the block coefficients, proving (2.2). The operators D and $\mathscr{A}$ commute, so $\delta^s = D^s \mathscr{A}^s$. Applying $D^s = \mathsf{U}^{ps}$ to (2.2) gives (2.3); terms with $Nn < ps$, equivalently $mn < s$, vanish. □

Definition 2.2 (Dwork-flat series)**.** Let $\mathcal{H}^{[p]}(\mathbb{Z}_p)$ be the subring supported in degrees divisible by p, and put

$$\mathscr{S}_p = \left\{ X \in \mathcal{H}^{[p]}(\mathbb{Z}_p) : \delta^s X \in p^s \mathcal{H}(\mathbb{Z}_p) \text{ for every } s \geq 0 \right\}. \tag{2.4}$$

Worked example 2.3 (Operator extraction for $N = 10$)**.** For $p = 5$, $m = 2$, $h = 2$,

$$\delta = \mathsf{U}^{25} - \mathsf{U}^5.$$

If $X = \sum x_n \gamma_{10n}$, then

$$\delta^2 X = \sum_{2n\geq 2} \Delta_2^2 x_n \gamma_{10n-10}.$$

Hence $\delta^2 X \in 25\mathcal{H}(\mathbb{Z}_5)$ implies $v_5(\Delta_2^2 x_n) \geq 2$ for every $n \geq 1$.

3. Ratio factorization and exact singular forms

Put

$$A_c(t) = t^{-c} H_{N,c}(t) = \sum_{n\geq 0} \frac{t^{Nn}}{(Nn+c)!}, \tag{3.1}$$

and, for $1 \leq c < N$,

$$R_c(t) = t\frac{H_{N,c-1}(t)}{H_{N,c}(t)} = \frac{A_{c-1}(t)}{A_c(t)}. \tag{3.2}$$

Proposition 3.1 (Factorization, Riccati equation, and recurrence)**.** *For $0 \leq j < N$,*

$$\boxed{F_{N,j} = F_{N,0} \prod_{c=1}^{j} R_c.} \tag{3.3}$$

For $c \geq 2$,

$$\boxed{tR_c' = R_c(1 + R_{c-1} - R_c).} \tag{3.4}$$

If

$$R_c(t) = \sum_{n\geq 0} r_n^{(c)} \frac{t^{Nn}}{(Nn)!},$$

then $r_0^{(c)} = c$ and

$$\boxed{\sum_{q=0}^{n} \binom{Nn+c}{Nq} r_q^{(c)} = Nn + c.} \tag{3.5}$$

Proof. Since $H_{N,c} = t^c A_c$, (1.3) gives $F_{N,c} = A_c^{-1}$. The product in (3.3) telescopes. Also $H_{N,c}' = H_{N,c-1}$, hence $R_c = tH_{N,c}'/H_{N,c}$. Differentiating and using $tH_{N,c-1}'/H_{N,c-1} = R_{c-1}$ gives

$$tR_c' = R_c + t^2 \frac{H_{N,c-2}}{H_{N,c}} - t^2 \left(\frac{H_{N,c-1}}{H_{N,c}}\right)^2 = R_c(1 + R_{c-1} - R_c).$$

Finally, $R_c H_{N,c} = tH_{N,c}'$. Comparing the coefficient of $t^{Nn+c}/(Nn+c)!$ gives (3.5). □

The Riccati propagation operator $\mathsf{E} + c$ has p-adic-unit eigenvalues on p-supported Hurwitz series precisely when $p \nmid c$. Thus the singular propagation indices are

$$\boxed{R_p, R_{2p}, \ldots, R_{(m-1)p}.} \tag{3.6}$$

Proposition 3.2 (Universal hypergeometric and gamma forms)**.** *Define*

$$\Phi_p(w) = \sum_{k\geq 0} \frac{p^k w^k}{(pk)!}.$$

Then

$$\Phi_p(w) = {}_0F_{p-1}\left(\begin{matrix} - \\ \frac{1}{p}, \frac{2}{p}, \ldots, \frac{p-1}{p} \end{matrix}; \frac{w}{p^{p-1}}\right). \tag{3.7}$$

For $1 \leq a < m$, let

$$\Phi_{p;m,a}(w) = \frac{1}{m} \sum_{\ell=0}^{m-1} \omega_m^{-a\ell} \Phi_p(\omega_m^{\ell} w).$$

With $w = t^p/p$,

$$H_{mp,ap}(t) = \Phi_{p;m,a}(w), \qquad R_{ap}(t) = pw\frac{d}{dw}\log\Phi_{p;m,a}(w). \tag{3.8}$$

Moreover, with

$$G_{p,m,a}(x) = (ap)!\sum_{n\geq 0}\frac{x^n}{(Nn+ap)!},$$

and

$$\mathcal{B}_{p,m,a} = \left\{\frac{ap+1}{N},\dots,\frac{ap+N}{N}\right\}\setminus\{1\},$$

one has

$$G_{p,m,a}(x) = {}_0F_{N-1}\left(\begin{matrix}-\\ \mathcal{B}_{p,m,a}\end{matrix};\frac{x}{N^N}\right), \tag{3.9}$$

$$R_{ap}(t) = ap + Nx\frac{G'_{p,m,a}(x)}{G_{p,m,a}(x)}, \qquad x = t^N, \tag{3.10}$$

$$r_n^{(p,m,a)} = Nn(Nn)![x^n]\log G_{p,m,a}(x) \qquad (n\geq 1). \tag{3.11}$$

For Morita's normalization [2],

$$\Gamma_p(n) = (-1)^n\prod_{\substack{1\leq d<n\\ p\nmid d}} d \qquad (n\geq 1),$$

$$\Phi_p(w) = \sum_{k\geq 0}\frac{(-1)^{k+1}}{\Gamma_p(pk+1)}\frac{w^k}{k!}. \tag{3.12}$$

Proof. The coefficient identity

$$k!\prod_{r=1}^{p-1}\left(\frac{r}{p}\right)_k = \frac{(pk)!}{p^{pk}}$$

proves (3.7). The root-of-unity average defining $\Phi_{p;m,a}$ retains exactly the terms $k\equiv a \pmod m$; after $w = t^p/p$ these are precisely the terms of $H_{mp,ap}$. Taking the logarithmic derivative gives (3.8).

Likewise,

$$k!\prod_{\beta\in\mathcal{B}_{p,m,a}}(\beta)_k = \frac{(Nk+ap)!}{(ap)!N^{Nk}},$$

which proves (3.9). Since $H_{N,ap} = t^{ap}G_{p,m,a}(t^N)/(ap)!$, logarithmic differentiation proves (3.10) and coefficient extraction proves (3.11). Finally,

$$\Gamma_p(pk+1) = (-1)^{k+1}\frac{(pk)!}{p^k k!},$$

which is equivalent to (3.12). □

Worked example 3.3 (Closed form for $R_5^{(10)}$)**.** For $(p,m,a) = (5,2,1)$,

$$G(x) = 120\sum_{n\geq 0}\frac{x^n}{(10n+5)!} = {}_0F_9\left(\begin{matrix}-\\ \frac{3}{5},\frac{7}{10},\frac{4}{5},\frac{9}{10},\frac{11}{10},\frac{6}{5},\frac{13}{10},\frac{7}{5},\frac{3}{2}\end{matrix};\frac{x}{10^{10}}\right),$$

$$R_5^{(10)}(t) = 5 + 10x\frac{G'(x)}{G(x)}, \qquad x = t^{10}.$$

The recurrence gives

$$r_0 = 5, \qquad r_1 = \frac{10}{\binom{15}{5}} = \frac{10}{3003},$$

and

$$r_2 = \frac{20 - \binom{25}{10} r_1}{\binom{25}{20}} = -\frac{296614}{1450449}.$$

Exact diagnostic. The hypergeometric and gamma coefficient identities were checked for $p = 3, 5, 7, 11, 13$ and $0 \le k \le 6$: 70 checks. The logarithmic coefficient formula was checked in four singular families through $n = 5$: 20 checks. The first three $N = 10$ coefficients and the factorization at $N = 6, 10$ were also checked exactly. Every check passed.

Theorem 3.4 (Singular section theorem)**.** *For $1 \le a < m$, write*

$$R_{ap}(t) = \sum_{n \ge 0} r_n^{(p,m,a)} \frac{t^{Nn}}{(Nn)!}.$$

Then, for every $u \ge 1$ and $s, n \ge 0$,

$$\boxed{v_p\Big(\Delta_{uh}^s r_n^{(p,m,a)}\Big) \ge \min\{s, mn\}.} \tag{3.13}$$

Equivalently, $R_{ap} \in \mathscr{S}_p$.

4. Weighted Amice estimate for the singular sections

4.1. **The weighted Amice ring.**

Definition 4.1. Define

$$\mathscr{W}_p = \left\{ A(T) = \sum_{j \ge 0} a_j T^j \in \mathbb{Q}_p[[T]] : v_p(a_j) \ge -\left\lfloor \frac{j}{p} \right\rfloor \right\}. \tag{4.1}$$

Lemma 4.2 (Algebra and differential stability)**.** *The set $\mathscr{W}_p$ is a complete subring of $\mathbb{Q}_p[[T]]$. A member with unit constant term has its inverse in $\mathscr{W}_p$. Moreover, the operators*

$$\Theta_T = (1+T)\frac{d}{dT}, \qquad \mathcal{L}_T = \log(1+T)(1+T)\frac{d}{dT}$$

preserve $\mathscr{W}_p$.

Proof. Put $w(j) = \lfloor j/p \rfloor$. Since

$$w(i) + w(j-i) \le w(j),$$

the coefficient of T^j in a product of two members of $\mathscr{W}_p$ has valuation at least $-w(j)$. This proves closure under products; closure under sums and coefficientwise completeness are immediate. If $A(0)$ is a unit, the recursive coefficients of A^{-1} involve only products $a_i b_{j-i}$, and the same floor inequality proves the inverse assertion.

For $A = \sum a_j T^j$, the coefficient of T^j in $\Theta_T A$ is

$$(j+1)a_{j+1} + j a_j.$$

If $p \nmid j+1$, then $w(j+1) = w(j)$; if $p \mid j+1$, the extra factor $j+1$ compensates for the one-unit increase of the floor. Thus the first summand has valuation at least $-w(j)$, and the second does as well. Hence Θ_T preserves $\mathscr{W}_p$.

Finally,

$$\log(1+T) = \sum_{j \ge 1} (-1)^{j-1} \frac{T^j}{j}$$

belongs to $\mathscr{W}_p$. Indeed, if $v = v_p(j) \geq 1$, then $j \geq p^v$, and hence $\lfloor j/p \rfloor \geq p^{v-1} \geq v$. Product closure now proves the assertion for $\mathcal{L}_T$. □

4.2. **A cyclotomic trace bound.** Let $\mu_{p-1} \subset \mathbb{Z}_p^\times$ be the Teichmuller group.

Lemma 4.3 (Integer-valued binomial polynomials)**.** *For every $z \in \mathbb{Z}_p$ and every $j \geq 0$,*

$$\binom{z}{j} \in \mathbb{Z}_p.$$

Proof. Choose nonnegative integers $z_\nu \to z$ in $\mathbb{Z}_p$. The polynomial $X \mapsto \binom{X}{j}$ is continuous over $\mathbb{Q}_p$, while $\binom{z_\nu}{j} \in \mathbb{Z}$. Since $\mathbb{Z}_p$ is closed in $\mathbb{Q}_p$, the limit belongs to $\mathbb{Z}_p$. □

Lemma 4.4 (Cyclotomic trace lemma)**.** *For $\eta \in \mu_{p-1}$, define*

$$\Phi_\eta(T) = \frac{1}{p} \sum_{\xi^p=1} (1+T)^{\eta\xi}. \tag{4.2}$$

Then $\Phi_\eta(T) \in \mathscr{W}_p$.

Proof. Write

$$c_j = [T^j]\Phi_\eta(T) = \frac{1}{p} \sum_{\xi^p=1} \binom{\eta\xi}{j}, \qquad q = \left\lfloor \frac{j}{p} \right\rfloor.$$

We prove $p^q c_j \in \mathbb{Z}_p$.

When $q = 0$, the assertion is clear for $j = 0$. For $1 \leq j < p$, the polynomial $\binom{\eta X}{j}$ has degree less than p and zero constant term. Averaging its monomials over $X^p = 1$ therefore gives zero.

Assume $q \geq 1$ and put

$$B_\xi = p^q \binom{\eta\xi}{j}.$$

For $\xi = 1$, Lemma 4.3 gives $B_1 \in p\mathbb{Z}_p$.

Now fix a primitive pth root ξ and put $K = \mathbb{Q}_p(\xi)$ and $\pi = \xi - 1$. Normalize v_K by $v_K(\pi) = 1$, so $v_K(p) = p - 1$. Among the integers $0, 1, \ldots, j-1$, at least q are congruent to the residue of η modulo p. For each such integer d,

$$v_K(\eta\xi - d) = 1;$$

all other factors are integral. Hence

$$v_K\big((\eta\xi)_{\underline{j}}\big) \geq q.$$

Using

$$v_p(j!) = q + v_p(q!),$$

we obtain

$$\begin{aligned} v_K(B_\xi) &\geq q(p-1) + q - (p-1)v_p(j!) \\ &= q - (p-1)v_p(q!) = s_p(q) \geq 1, \end{aligned}$$

where $s_p(q)$ is the sum of the base-p digits of q. Thus $B_\xi \in \pi\mathcal{O}_K$. Every conjugate also lies in this ideal, and therefore

$$\sum_{\substack{\xi^p=1 \\ \xi\neq 1}} B_\xi = \mathrm{Tr}_{K/\mathbb{Q}_p}(B_\xi) \in \pi\mathcal{O}_K \cap \mathbb{Q}_p = p\mathbb{Z}_p.$$

Together with $B_1 \in p\mathbb{Z}_p$, this proves $p^q c_j = p^{-1} \sum_{\xi^p=1} B_\xi \in \mathbb{Z}_p$. □

Worked example 4.5 (The sharp first trace coefficient at $p = 5$)**.** For $\eta = 1$ and $j = 5$, only the top monomial of $\binom{X}{5}$ survives the average over $X^5 = 1$. Hence

$$[T^5]\Phi_1(T) = \frac{1}{5!} = \frac{1}{120}, \qquad v_5\Big([T^5]\Phi_1\Big) = -1 = -\left\lfloor \frac{5}{5} \right\rfloor.$$

For comparison, exact expansion gives

$$[T^{10}]\Phi_1(T) = -\frac{67331}{907200}, \qquad v_5 = -2 = -\left\lfloor \frac{10}{5} \right\rfloor.$$

Thus the weight $-\lfloor j/p \rfloor$ is attained and cannot in general be replaced by coefficientwise integrality.

4.3. **The weighted bound for a singular ratio.**

Lemma 4.6. *For $1 \le a < m$,*

$$H_{N,ap}(\log(1+T)) \in \mathscr{W}_p.$$

Proof. For $t = \log(1+T)$, Lemma 4.4 gives

$$\begin{aligned}\Phi_\eta(T) &= \frac{1}{p}\sum_{\xi^p=1} e^{\eta\xi t} \\ &= \sum_{q\ge 0} \eta^{pq}\frac{t^{pq}}{(pq)!} = \sum_{q\ge 0}\eta^q \frac{t^{pq}}{(pq)!},\end{aligned}$$

because $\eta^p = \eta$. Averaging over $\eta^m = 1$ with weight η^{-a} selects $q \equiv a \pmod m$ and yields

$$H_{N,ap}(t) = \frac{1}{m}\sum_{\eta^m=1} \eta^{-a}\Phi_\eta(T).$$

Since $p \nmid m$, the result follows. □

Proposition 4.7 (Weighted Amice estimate)**.** *For $1 \le a < m$,*

$$R_{ap}(\log(1+T)) \in \mathscr{W}_p.$$

Equivalently, if $R_{ap}(\log(1+T)) = \sum_{j\ge 0}\alpha_j T^j$, then

$$v_p(\alpha_j) \ge -\left\lfloor \frac{j}{p} \right\rfloor. \tag{4.3}$$

Proof. Put $c = ap$ and

$$U(T) = c!T^{-c}H_{N,c}(\log(1+T)).$$

The series U has constant coefficient one. Since $a < p$,

$$v_p(c!) = a, \qquad \left\lfloor \frac{j+c}{p} \right\rfloor = a + \left\lfloor \frac{j}{p} \right\rfloor.$$

Thus Lemma 4.6 gives $U \in \mathscr{W}_p$, and Lemma 4.2 gives $U^{-1} \in \mathscr{W}_p$.

The operator $t\,d/dt$ becomes $\mathcal{L}_T$ under $t = \log(1+T)$. Since

$$H_{N,c}(\log(1+T)) = \frac{T^c}{c!}U(T),$$

we have

$$R_c(\log(1+T)) = c\frac{(1+T)\log(1+T)}{T} + \frac{\mathcal{L}_T U}{U}. \tag{4.4}$$

The second term belongs to $\mathscr{W}_p$. The first term does as well. Indeed, its constant coefficient is c, while for $j \ge 1$ its T^j coefficient is

$$c\frac{(-1)^{j+1}}{j(j+1)}.$$

At most one of $j, j+1$ is divisible by p. If $v = v_p(j(j+1)) \geq 1$, then $p^v \leq j+1$, so

$$1 + \left\lfloor \frac{j}{p} \right\rfloor \geq p^{v-1} \geq v.$$

Thus $v_p(j(j+1)) \leq 1 + \lfloor j/p \rfloor$. Since $v_p(c) = 1$, the required lower bound follows. □

Worked example 4.8 (The $N = 10$ singular section)**.** For $(p, m, a) = (5, 2, 1)$,

$$R_5^{(10)}(\log(1+T)) = \sum_{j \geq 0} \alpha_j T^j.$$

The first nonconstant coefficient occurs at $j = 10$ and equals

$$\alpha_{10} = \frac{r_1}{10!} = \frac{1}{1089728640}, \qquad v_5(\alpha_{10}) = -1 \geq -2.$$

Two later exact values are

$$\alpha_{20} = \frac{41924625638075119}{80200006474040893440000}, \quad v_5(\alpha_{20}) = -4,$$

$$\alpha_{30} = \frac{22880067374952906684621513641 53}{84469678848467408246466517401 6000000}, \quad v_5(\alpha_{30}) = -6.$$

Both attain the predicted lower bound $-\lfloor j/5 \rfloor$.

Exact diagnostic. The cyclotomic trace inequality was checked exactly for $p = 3, 5, 7, 11$, $\eta = \pm 1$, and $0 \leq j \leq 4p^2$: 1,640 rational checks. The bound $v_p(\alpha_j) \geq -\lfloor j/p \rfloor$ was checked for five singular families through $j = 36$: 185 rational checks. Every check passed.

5. Dual Mahler filtration and the singular theorem

5.1. The dual Mahler filtration.

Definition 5.1. For a polynomial $P \in \mathbb{Q}_p[x]$, put

$$\Delta P(x) = P(x+1) - P(x).$$

Define

$$\mathscr{P}_p = \left\{ P \in \mathbb{Q}_p[x] : v_p(\Delta^j P(0)) \geq \left\lfloor \frac{j}{p} \right\rfloor \text{ for all } j \geq 0 \right\}. \tag{5.1}$$

Lemma 5.2 (Mahler ring)**.** *The set $\mathscr{P}_p$ is a subring of $\mathbb{Q}_p[x]$, and $\mathbb{Z}_p[x] \subseteq \mathscr{P}_p$.*

Proof. Use the Mahler basis $\binom{x}{j}$; see, for example, Mahler [1]. The product formula is

$$\binom{x}{i}\binom{x}{j} = \sum_{k=\max(i,j)}^{i+j} C_{i,j,k} \binom{x}{k}, \tag{5.2}$$

where

$$C_{i,j,k} = \frac{k!}{(k-i)!(k-j)!(i+j-k)!} = \binom{k}{i}\binom{i}{i+j-k}.$$

It is enough to prove

$$v_p(C_{i,j,k}) + \left\lfloor \frac{i}{p} \right\rfloor + \left\lfloor \frac{j}{p} \right\rfloor \geq \left\lfloor \frac{k}{p} \right\rfloor. \tag{5.3}$$

The right floor exceeds the sum of the two left floors by at most one, because $k \leq i + j$. Suppose that the excess equals one. Write

$$i = pa + r, \qquad j = pb + s, \qquad k = p(a+b+1) + t,$$

with $0 \leq r, s, t < p$. The inequality $k \leq i + j$ gives $t \leq r + s - p < r$. Lucas's theorem then gives $\binom{k}{i} \equiv 0 \pmod{p}$, so $p \mid C_{i,j,k}$ and (5.3) follows. This proves multiplicative closure.

Finally,

$$\Delta^j x^d\big|_{x=0} = j!S(d,j),$$

where $S(d,j)$ is a Stirling number of the second kind. Hence every Mahler coefficient of a polynomial in $\mathbb{Z}_p[x]$ has valuation at least $v_p(j!) \geq \lfloor j/p \rfloor$. □

Worked example 5.3 (A carry in the Mahler product at $p = 5$)**.** For $i = j = 4$,

$$\binom{x}{4}^2 = \sum_{k=4}^{8} C_{4,4,k} \binom{x}{k},$$

with

k	4	5	6	7	8
$C_{4,4,k}$	1	20	90	140	70
$v_5(C_{4,4,k})$	0	1	1	1	1

The terms $k \geq 5$ cross one 5-adic weight boundary, and each coefficient acquires the required factor 5.

5.2. **A null polynomial in the Mahler filtration.**

Lemma 5.4 (Stirling congruence)**.** *For $0 \leq j < p^2$,*

$$S(p^2, j) \equiv S(p, j) \pmod{p}.$$

Proof. Let the cyclic group C_{p^r} act by translation on the set of set partitions of its underlying p^r elements. Every orbit not fixed by the whole group has cardinality divisible by p. A partition fixed by translation is a translation-invariant equivalence relation, hence its blocks are the cosets of a subgroup of C_{p^r}. There is exactly one such subgroup for every index $1, p, \ldots, p^r$. It follows that

$$S(p^r, j) \equiv \begin{cases} 1 \pmod{p}, & j \in \{1, p, \ldots, p^r\}, \\ 0 \pmod{p}, & \text{otherwise.} \end{cases}$$

Comparing $r = 2$ and $r = 1$, with $S(p, j) = 0$ for $j > p$, proves the claim for $j < p^2$. □

Lemma 5.5 (Null-polynomial element)**.** *The polynomial*

$$\mathcal{N}_p(x) = \frac{(x^p - x)^p}{p} \tag{5.4}$$

belongs to $\mathscr{P}_p$.

Proof. The binomial theorem gives

$$\mathcal{N}_p(x) = \frac{x^{p^2} - x^p}{p} + Q(x), \qquad Q(x) \in \mathbb{Z}[x].$$

By Lemma 5.2, it is enough to treat the first term. Its jth Mahler coefficient is

$$\frac{j!}{p}(S(p^2, j) - S(p, j)).$$

For $j < p^2$, Lemma 5.4 makes this an element of $j!\mathbb{Z}_p$, whose valuation is at least $\lfloor j/p \rfloor$. At $j = p^2$, the coefficient is $p^2!/p$, and

$$v_p(p^2!/p) = p = \left\lfloor \frac{p^2}{p} \right\rfloor.$$

Higher Mahler coefficients vanish. □

Worked example 5.6 (Mahler weights of $\mathcal{N}_5$)**.** For

$$\mathcal{N}_5(x) = \frac{(x^5 - x)^5}{5},$$

the exact valuations of selected Mahler coefficients $\Delta^j \mathcal{N}_5(0)$ are

j	5	10	15	20	25
$v_5(\Delta^j \mathcal{N}_5(0))$	4	5	7	5	5
$\lfloor j/5 \rfloor$	1	2	3	4	5

which illustrates the inclusion $\mathcal{N}_5 \in \mathscr{P}_5$.

5.3. **The decisive polynomial lemma.** Put $L = p(p-1)$.

Lemma 5.7 (Weighted null-polynomial lemma)**.** *For all $b, s \geq 0$ and every $u \geq 1$,*

$$\boxed{p^{-\min\{b,s\}} x^{pb} (x^{uL} - 1)^s \in \mathscr{P}_p.} \tag{5.5}$$

Proof. First take $u = 1$ and define

$$C_p(x) = \frac{x^L - 1 - (x^{p-1} - 1)^p}{p} \in \mathbb{Z}[x].$$

The integrality follows because all intermediate binomial coefficients in the pth row are divisible by p.

Assume $b \geq s$. Since

$$x^{p-1} - 1 = \frac{x^p - x}{x},$$

expansion of $x^L - 1 = (x^{p-1} - 1)^p + pC_p(x)$ gives

$$\frac{x^{pb}(x^L - 1)^s}{p^s} = \sum_{i=0}^{s} \binom{s}{i} x^{p(b-i)} C_p(x)^{s-i} \mathcal{N}_p(x)^i. \tag{5.6}$$

Every factor on the right belongs to $\mathscr{P}_p$ by Lemmas 5.2 and 5.5; hence so does the left side.

If $b < s$, factor

$$\frac{x^{pb}(x^L - 1)^s}{p^b} = \left(\frac{x^{pb}(x^L - 1)^b}{p^b} \right) (x^L - 1)^{s-b}$$

and use the case $b = s$. Finally,

$$x^{uL} - 1 = (x^L - 1)(1 + x^L + \cdots + x^{(u-1)L}),$$

so the general case follows by multiplying by an integral polynomial. □

Worked example 5.8 (Boundary polynomial for $N = 10$)**.** Take $p = 5$, $b = s = 2$, $u = 1$, and $L = 20$. Then

$$\frac{x^{10}(x^{20} - 1)^2}{25} = x^{10} C_5(x)^2 + 2x^5 C_5(x) \mathcal{N}_5(x) + \mathcal{N}_5(x)^2 \in \mathscr{P}_5.$$

Selected Mahler valuations of the left side are

j	5	10	15	20	25	30	35	40	45	50
$v_5(\Delta^j P(0))$	6	5	5	5	8	8	11	9	11	10
$\lfloor j/5 \rfloor$	1	2	3	4	5	6	7	8	9	10

where $P = x^{10}(x^{20} - 1)^2/25$.

5.4. **The pairing and completion of the singular proof.** Write

$$R_{ap}(\log(1+T)) = \sum_{j\ge 0} \alpha_j T^j.$$

Define a linear functional on polynomials by

$$\mu\left(\binom{x}{j}\right) = \alpha_j. \tag{5.7}$$

Lemma 5.9 (Integral pairing)**.** *If $P \in \mathscr{P}_p$, then $\mu(P) \in \mathbb{Z}_p$. Moreover,*

$$\mu(x^d) = [t^d/d!]R_{ap}(t). \tag{5.8}$$

Proof. The Mahler expansion of a polynomial is finite:

$$P(x) = \sum_j \Delta^j P(0)\binom{x}{j}.$$

The defining bounds for $\mathscr{P}_p$ and (4.3) show that every summand in $\mu(P)$ is integral.

For the moment identity, substitute $T = e^t - 1$:

$$R_{ap}(t) = \sum_j \alpha_j (e^t-1)^j.$$

Since

$$(e^t-1)^j = j!\sum_{d\ge j} S(d,j)\frac{t^d}{d!}, \qquad x^d = \sum_j j!S(d,j)\binom{x}{j},$$

coefficient comparison gives (5.8). □

Proof of Theorem 3.4. By (5.8),

$$r_n^{(p,m,a)} = \mu(x^{Nn}).$$

Therefore

$$\Delta_{uh}^s r_n^{(p,m,a)} = \mu\Big(x^{Nn}(x^{Nuh}-1)^s\Big).$$

Now

$$Nn = p(mn), \qquad Nuh = up(p-1) = uL.$$

By Lemma 5.7,

$$p^{-\min\{mn,s\}} x^{Nn}(x^{Nuh}-1)^s \in \mathscr{P}_p.$$

Applying Lemma 5.9 proves (3.13). Taking $u = 1$ and using (2.3) gives $R_{ap} \in \mathscr{S}_p$. □

Worked example 5.10 (A complete singular difference)**.** For $R_5^{(10)}(t) = \sum r_n t^{10n}/(10n)!$,

$$\Delta_2^2 r_1 = r_5 - 2r_3 + r_1 = \frac{24606419809197847119397525}{12490016994}.$$

The denominator is a 5-adic unit and the numerator has valuation 2, so

$$v_5(\Delta_2^2 r_1) = 2 = \min\{2, 2\cdot 1\}.$$

Exact diagnostic. The Mahler product inequality was checked for $p = 3, 5, 7, 11$ and $0 \le i, j < 40$: 88,560 checks. The Stirling and null-polynomial lemmas contributed 412 checks; the weighted polynomial lemma was checked for $p = 3, 5, 7$, $0 \le b, s \le 3$, and $u = 1, 2$: 5,712 checks. The singular finite-difference theorem was checked in every singular family for primes through 19, with $u = 1, 2$, $0 \le n \le 4$, and $0 \le s \le \min(4, mn)$: 4,496 checks. Every check passed.

6. Dwork-flat algebra and nonsingular propagation

6.1. **Closure of the Dwork-flat class.**

Lemma 6.1 (Frobenius defect)**.** *For $X, Y \in \mathcal{H}(\mathbb{Z}_p)$,*

$$\delta(XY) = \delta(X)Y + X\delta(Y) + p\mathcal{B}_p(X, Y),$$

where

$$\mathcal{B}_p(X,Y) = \sum_{a=1}^{p^2-1} \frac{1}{p}\binom{p^2}{a}(\mathsf{U}^a X)(\mathsf{U}^{p^2-a}Y) - \sum_{a=1}^{p-1} \frac{1}{p}\binom{p}{a}(\mathsf{U}^a X)(\mathsf{U}^{p-a}Y).$$

Every coefficient in this bilinear expression is p-integral.

Proof. Apply the Leibniz rule to $\mathsf{U}^{p^2}(XY)$ and $\mathsf{U}^p(XY)$ and subtract. The endpoint terms give $\delta(X)Y + X\delta(Y)$. Every intermediate binomial coefficient in a prime-power row is divisible by p. □

Lemma 6.2 (Stable algebra)**.** *The class $\mathscr{S}_p$ is closed under sums and products. If $X \in \mathscr{S}_p$ has a unit constant coefficient, then $X^{-1} \in \mathscr{S}_p$. It is also closed under U^p and under coefficientwise t-adic limits.*

Proof. Let $X_1, \dots, X_r \in \mathscr{S}_p$, and let $\mathcal{R}$ be the U-stable $\mathbb{Z}_p$-subalgebra of $\mathcal{H}(\mathbb{Z}_p)$ generated by

$$\mathsf{U}^a\big(p^{-s}\delta^s X_i\big) \qquad (a, s \geq 0,\ 1 \leq i \leq r).$$

All generators are integral. Since δ commutes with U,

$$\delta\big(\mathsf{U}^a p^{-s}\delta^s X_i\big) = p\,\mathsf{U}^a\Big(p^{-(s+1)}\delta^{s+1}X_i\Big) \in p\mathcal{R}.$$

If $A, B \in \mathcal{R}$ satisfy $\delta(A), \delta(B) \in p\mathcal{R}$, then Lemma 6.1 gives

$$\delta(AB) = \delta(A)B + A\delta(B) + p\mathcal{B}_p(A,B) \in p\mathcal{R},$$

because $\mathcal{R}$ is U-stable. Induction on algebraic expressions yields $\delta(\mathcal{R}) \subseteq p\mathcal{R}$, and then

$$\delta^s(\mathcal{R}) \subseteq p^s\mathcal{R} \qquad (s \geq 0).$$

This proves closure under sums and products.

Suppose now that $X \in \mathscr{S}_p$ has unit constant coefficient. The usual coefficient recursion gives $X^{-1} \in \mathcal{H}^{[p]}(\mathbb{Z}_p)$. Enlarge $\mathcal{R}$ to the U-stable algebra $\mathcal{R}'$ generated also by all U-derivatives of X^{-1}. Applying Lemma 6.1 to $XX^{-1} = 1$ gives

$$X\delta(X^{-1}) = -\delta(X)X^{-1} - p\mathcal{B}_p(X, X^{-1}) \in p\mathcal{R}'.$$

Multiplication by X^{-1} yields $\delta(X^{-1}) \in p\mathcal{R}'$. Since δ commutes with U, the same is true for all added generators. Repeating the preceding algebra argument gives $\delta^s(X^{-1}) \in p^s\mathcal{H}(\mathbb{Z}_p)$.

Closure under U^p follows from $[\delta, \mathsf{U}] = 0$. For fixed s, δ^s is a finite linear combination of derivatives of order at most $p^2 s$; it is therefore continuous for the t-adic topology. Divisibility passes coefficientwise to t-adic limits. □

6.2. **The unit starting interval.**

Lemma 6.3 (Unit interval)**.** *For $0 \le c < p$, the series A_c, $F_{N,c}$, and, when $c \ge 1$, R_c belong to $\mathscr{S}_p$.*

Proof. Normalize

$$G_c(t) = c!A_c(t) = \sum_{n\ge 0} g_c(n)\gamma_{Nn}, \qquad g_c(n) = \binom{Nn+c}{c}^{-1}.$$

For $x \in \mathbb{Z}_p$,

$$g_c(x) = \prod_{d=1}^{c} \left(1 + \frac{N}{d}x\right)^{-1} = \sum_{q\ge 0} b_q x^q, \qquad b_q \in p^q\mathbb{Z}_p,$$

because every $1 \le d \le c < p$ is a unit and $v_p(N) = 1$. Since $v_p(b_q) \ge q$, this expansion converges uniformly for $x \in \mathbb{Z}_p$. The operator Δ_h^s annihilates polynomials of degree less than s and preserves $\mathbb{Z}_p[x]$; termwise application is therefore valid and gives

$$\Delta_h^s g_c(n) \in p^s\mathbb{Z}_p \qquad (s, n \ge 0).$$

Equation (2.3) gives $\delta^s G_c \in p^s\mathcal{H}(\mathbb{Z}_p)$, so $G_c \in \mathscr{S}_p$. Since $c!$ is a unit, $A_c \in \mathscr{S}_p$. Finally,

$$F_{N,c} = A_c^{-1}, \qquad R_c = A_{c-1}A_c^{-1},$$

and Lemma 6.2 applies. □

Worked example 6.4 (Unit interval at $N = 10$)**.** For $c = 1$,

$$g_1(x) = \binom{10x+1}{1}^{-1} = \frac{1}{1+10x}.$$

At step $h = 2$,

$$\Delta_2^2 g_1(0) = g_1(4) - 2g_1(2) + g_1(0) = \frac{1}{41} - \frac{2}{21} + 1 = \frac{800}{861},$$

so $v_5(\Delta_2^2 g_1(0)) = 2$.

6.3. **Nonsingular propagation.**

Lemma 6.5 (Integral Euler-operator inverse)**.** *Suppose $p \nmid c$ and put*

$$\Lambda_c = \mathsf{E} + c.$$

On $\mathcal{H}^{[p]}(\mathbb{Z}_p)$, the operator Λ_c is an integral bijection. If $X \in \mathscr{S}_p$ and $Y = \Lambda_c^{-1}X$, then $Y \in \mathscr{S}_p$.

Proof. For $p \mid d$,

$$\Lambda_c\gamma_d = (d + c)\gamma_d,$$

and $d + c$ is a p-adic unit. This proves integral invertibility.

Set

$$C = p\mathsf{U}^{p^2} - \mathsf{U}^p.$$

A direct computation from $[\mathsf{U}^r, \mathsf{E}] = r\mathsf{U}^r$ gives

$$[\delta, \Lambda_c] = pC.$$

Since C commutes with δ,

$$[\delta^s, \Lambda_c] = spC\delta^{s-1}. \tag{6.1}$$

Assume inductively that $\delta^{s-1}Y \in p^{s-1}\mathcal{H}(\mathbb{Z}_p)$. From $\Lambda_c Y = X$ and (6.1),

$$\Lambda_c\delta^s Y = \delta^s X - spC\delta^{s-1}Y \in p^s\mathcal{H}(\mathbb{Z}_p).$$

The right side remains supported in degrees divisible by p, so the integral inverse of Λ_c proves $\delta^s Y \in p^s\mathcal{H}(\mathbb{Z}_p)$. □

Proposition 6.6 (Nonsingular propagation)**.** *Let $2 \le c < N$ and suppose $p \nmid c$. If $R_{c-1} \in \mathscr{S}_p$, then $R_c \in \mathscr{S}_p$.*

Proof. Write

$$R_{c-1} = (c-1) + q, \qquad R_c = c + y,$$

where q, y have positive t-adic order at least N. Equation (3.4) is equivalent to

$$\Lambda_c y = cq + yq - y^2. \tag{6.2}$$

Consider the map

$$\Phi(y) = \Lambda_c^{-1}(cq + yq - y^2)$$

on $t^N\mathcal{H}^{[p]}(\mathbb{Z}_p)$. If $y - z$ has order d, then

$$\Phi(y) - \Phi(z) = \Lambda_c^{-1}\big((y-z)(q-y-z)\big)$$

has order at least $d+N$. Hence iteration from $y_0 = 0$ converges t-adically to a solution $y \in t^N\mathcal{H}^{[p]}(\mathbb{Z}_p)$. This solution is unique even in $t^N\mathbb{Q}_p[[t^N]]$: if y and z are two solutions and d is the least degree of $y - z$, then the least degree of $\Lambda_c(y-z)$ is d, whereas the least degree of $(y-z)(q-y-z)$ is at least $d+N$, a contradiction. Since the formal ratio $R_c - c$ satisfies the same equation, it equals y.

By hypothesis $q \in \mathscr{S}_p$. The class $\mathscr{S}_p$ is closed under products and sums by Lemma 6.2, and Λ_c^{-1} preserves it by Lemma 6.5. Thus every iterate y_ν lies in $\mathscr{S}_p$, and so does its limit. Therefore $R_c = c + y \in \mathscr{S}_p$. □

Worked example 6.7 (Propagation from R_5 to R_6 at $N = 10$)**.** Write

$$R_5 = 5 + q_1\gamma_{10} + O(t^{20}), \qquad R_6 = 6 + y_1\gamma_{10} + O(t^{20}),$$

where $q_1 = 10/3003$. The degree-10 part of $\Lambda_6 y = 6q + yq - y^2$ is

$$(10+6)y_1 = 6q_1.$$

Therefore

$$y_1 = \frac{6}{16}\frac{10}{3003} = \frac{5}{4004},$$

which agrees with the direct ratio recurrence.

Exact diagnostic. The Frobenius-defect divisibilities and the commutator formula were checked in 278 exact integer tests. The Riccati identity was checked for $N = 10, 20, 21$, every $2 \le c < N$, and four nonconstant coefficients: 180 checks. Ratio factorization was checked for every j at $N = 6, 10$ through block index 4: 70 checks. Every check passed.

Theorem 6.8 (All ratios are Dwork-flat)**.** *For every $1 \le c < N$, $R_c \in \mathscr{S}_p$. Moreover, $F_{N,0} \in \mathscr{S}_p$, and therefore*

$$F_{N,j} = F_{N,0}\prod_{c=1}^{j} R_c \in \mathscr{S}_p \qquad (0 \le j < N).$$

6.4. **All ratios and coefficient extraction.**

Proof of Theorem 6.8. The ratios $R_1, \dots, R_{p-1}$ and $F_{N,0}$ are supplied by Lemma 6.3. For each $1 \le a < m$, the singular ratio R_{ap} is supplied by Theorem 3.4. Starting from each R_{ap}, repeated use of Proposition 6.6 fills the interval

$$R_{ap+1}, R_{ap+2}, \dots, R_{(a+1)p-1}.$$

Thus every R_c, $1 \le c < N$, belongs to $\mathscr{S}_p$. Finally, (3.3) and product closure give every $F_{N,j}$. □

Worked example 6.9 (Reconstruction of $F_{10,7}$)**.** The factorization is

$$F_{10,7} = F_{10,4}R_5R_6R_7.$$

The constant coefficient is $24 \cdot 5 \cdot 6 \cdot 7 = 7!$. Using

$$[\gamma_{10}]F_{10,4} = -\frac{24}{1001}, \quad [\gamma_{10}]R_5 = \frac{10}{3003}, \quad [\gamma_{10}]R_6 = \frac{5}{4004}, \quad [\gamma_{10}]R_7 = \frac{5}{9724},$$

one obtains

$$\begin{aligned} E_{10}^{(10,7)} = & -\frac{24}{1001}(5)(6)(7) + 24\frac{10}{3003}(6)(7) \\ & + 24(5)\frac{5}{4004}(7) + 24(5)(6)\frac{5}{9724} = -\frac{630}{2431}. \end{aligned}$$

Proof of Theorem 1.4. Fix j, k, n and put $s = \min\{k, mn\}$. By Theorem 6.8, $F_{N,j} \in \mathscr{S}_p$. Since $\mathscr{A}$ is a polynomial in U^p, it preserves $\mathscr{S}_p$. The operators D and $\mathscr{A}$ commute, so

$$D^s\mathscr{A}^k F_{N,j} = (D\mathscr{A})^s\mathscr{A}^{k-s}F_{N,j} = \delta^s\mathscr{A}^{k-s}F_{N,j} \in p^s\mathcal{H}(\mathbb{Z}_p). \tag{6.3}$$

By (2.2), the coefficient of γ_{Nn-ps} in the left side is $\Delta_h^k e_n^{(N,j)}$. The index is nonnegative because $Nn - ps = p(mn - s) \geq 0$. This proves (1.6) and, with $k = 0$, the p-integrality of the coefficients.

For a multiple step, let $T_h = 1 + \Delta_h$ be translation by h. Then

$$\Delta_{uh} = T_h^u - 1 = \Delta_h(1 + T_h + \cdots + T_h^{u-1}).$$

Thus Δ_{uh}^k is an integral linear combination of Δ_h^ℓ with $\ell \geq k$, and (1.7) follows from (1.6). □

Proof of Theorem 1.5. Newton's formula gives

$$e_{n+hq} - e_n = \sum_{k=1}^{q} \binom{q}{k} \Delta_h^k e_n.$$

For $1 \leq k \leq q$,

$$v_p\binom{q}{k} \geq v_p(q) - v_p(k).$$

If $k \leq mn$, then Theorem 1.4 gives

$$v_p\left(\binom{q}{k}\Delta_h^k e_n\right) \geq v_p(q) - v_p(k) + k \geq v_p(q) + 1,$$

because $k - v_p(k) \geq 1$. If $k > mn$, the same term has valuation at least mn. Every summand therefore has valuation at least $\min\{mn, 1 + v_p(q)\}$, proving (1.8). Take $q = up^{r-1}$ and use $Nh = p(p-1)$ to obtain (1.9) and (1.10). □

Worked example 6.10 (The proved $N = 10$, $j = 7$ layer)**.** Here $p = 5$, $m = 2$, $h = 2$, and $r = 3$. For $n = 2$ and $q = 25$,

$$v_5(e_{2+2q} - e_2) \geq \min\{4, 1 + v_5(25)\} = 3.$$

Thus

$$E_{520}^{(10,7)} \equiv E_{20}^{(10,7)} \pmod{125}.$$

The defining recurrence gives

$$E_{20}^{(10,7)} = \frac{412013}{167739} \equiv 67 \pmod{125},$$

and exact computation gives $E_{520}^{(10,7)} \equiv 67 \pmod{125}$.

Worked example 6.11 (The $N = 6$, $j = 3$ block modulo 27)**.** For $p = 3$, $m = 2$, $h = 1$, and $r = 3$, the theorem gives the block shift $h3^{r-1} = 9$ from $n \geq \lceil 3/2 \rceil = 2$. Exact residues are

n	2	3	4	5	6	7	8	9	10
$E_{6n}^{(6,3)} \bmod 27$	10	13	16	19	22	25	1	4	7

and the same row occurs for $11 \leq n \leq 19$.

Exact diagnostic. The global bound was checked for every residue class $0 \leq j < mp$, all divisors $m \mid p-1$, primes $p = 3, 5, 7, 11, 13$, $0 \leq n \leq 3$, and $0 \leq k \leq 4$: 13,800 checks. The local modulus was checked for primes through 11 and $1 \leq q \leq 8$: 10,432 checks. Every check passed.

7. Corollaries and recovered examples

Corollary 7.1 (Generalized Euler and Lehmer numbers)**.** *For $j = 0$,*

$$v_p\Big(\Delta_{uh}^k E_{Nn}^{(N,0)}\Big) \geq \min\{k, mn\}.$$

For $N = 3$ this gives the Lehmer-number estimate

$$v_3\Big(\Delta_{2u}^k W_{3n}\Big) \geq \min\{k, n\}.$$

Corollary 7.2 (Bernoulli subsequence at $N = 6$)**.** *The identity*

$$t\frac{H_{6,2}(t)}{H_{6,3}(t)} = 3\sum_{n\geq 0} B_{6n}\frac{t^{6n}}{(6n)!}$$

implies

$$\boxed{v_3(\Delta_u^s(3B_{6n})) \geq \min\{s, 2n\}.}$$

Proof. Nishibuchi's sine factorization of $H_{6,3}$ [3, Proposition 4.1] gives

$$t\frac{H'_{6,3}(t)}{H_{6,3}(t)} = \sum_{\alpha\in\mathcal{A}} \alpha t\cot(\alpha t), \quad \mathcal{A} = \left\{\frac{i}{2}, \frac{\sqrt{3}+i}{4}, \frac{\sqrt{3}-i}{4}\right\}.$$

The numbers $4\alpha^2$ are the three roots of $z^3 + 1$. Substitution of

$$z\cot z = \sum_{q\geq 0}(-1)^q 2^{2q} B_{2q}\frac{z^{2q}}{(2q)!}$$

cancels all indices except $2q = 6n$ and leaves the coefficient $3B_{6n}$. Apply Theorem 3.4 with $(p, m, a) = (3, 2, 1)$. □

Corollary 7.3 (The complete $N = 10$ singular range)**.** *For the sequence in Example 3.3,*

$$\boxed{v_5(\Delta_{2u}^s r_n) \geq \min\{s, 2n\}.}$$

The factorization and propagation theorem therefore cover every class $j = 5, 6, 7, 8, 9$ at $N = 10$.

Corollary 7.4 (Finite Hurwitz products)**.** *Let*

$$\prod_{\nu=1}^{d} F_{N,j_\nu}(t) = \sum_{n\geq 0} c_n\frac{t^{Nn}}{(Nn)!}.$$

Then

$$v_p(\Delta_{uh}^k c_n) \geq \min\{k, mn\},$$

and

$$v_p(c_{n+hq} - c_n) \geq \min\{mn, 1 + v_p(q)\}.$$

Thus the theorem also applies to all finite binomial-convolution sequences generated by the $F_{N,j}$.

Proof. Use product closure of $\mathscr{S}_p$, followed by the same coefficient extraction and Newton expansion used for Theorems 1.4 and 1.5. □

Table 1. Representative periodicity layers supplied by the theorem.

(N, j)	p	m	r	h	$n_0 = \lceil r/m \rceil$	period
$(6, 1)$	3	2	2	1	1	18
$(6, 3)$	3	2	5	1	3	486
$(10, 4)$	5	2	2	2	1	100
$(10, 7)$	5	2	3	2	2	500
$(20, 13)$	5	4	3	1	1	500
$(21, 8)$	7	3	2	2	1	294
$(42, 9)$	7	6	3	1	1	2058

8. Verification ledger

All computations use exact integers or reduced rational numbers and are independent of the proofs. The accompanying script reports

125,858 exact checks, all passed.

Proof interface	Checks
Universal hypergeometric and gamma coefficients	70
Cyclotomic trace bound on rational branches	1,640
Stirling and null-polynomial lemmas	412
Frobenius defect and commutator	278
Initial $N = 10$ coefficients	3
Hypergeometric logarithmic derivative	20
Weighted Amice coefficients	185
Singular finite differences	4,496
All residue classes	13,800
Strong local modulus	10,432
Ratio factorization	70
Mahler product weights	88,560
Weighted null-polynomial lemma	5,712
Riccati ratio identity	180
Total	125,858

The tests do not replace any argument. They verify the coefficient normalizations, signs, shifts, and valuations at every interface of the proof.

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